RANDOM FOREST PROXIMITIES

BENEFITS OF RANDOM FORESTS

- Classification and regression
- Mixed variable types
- Estimate of generalization error
- Trivially parallelizable
- Scale invariant
- Insensitive to outliers in predictor space
- Model non-linear relationships
- Natural produce a measure of similarity

RANDOM FOREST PROXIMITIES

"[Proximities] are one of the most useful tools in random forests."

--Leo Breiman

Original description:

"The proximities originally formed a NxN matrix. After a tree is grown, put all of the data, both training and oob, down the tree. If cases k and n are in the same terminal node increase their proximity by one. At the end, normalize the proximities by dividing by the number of trees."

https://www.stat.berkeley.edu/~breiman/RandomForests/cc_home.htm#:~:text=few%20data%20sets.-,Proximities,by%20the%20number%20of%20trees.

PROXIMITIES IN A NUTSHELL

“Proximities don’t just measure the similarity of the variables---they also take into account the importance of the variables.”

“Two cases that have quite different predictor variables might have large proximity if they differ only on variables that are not important.”

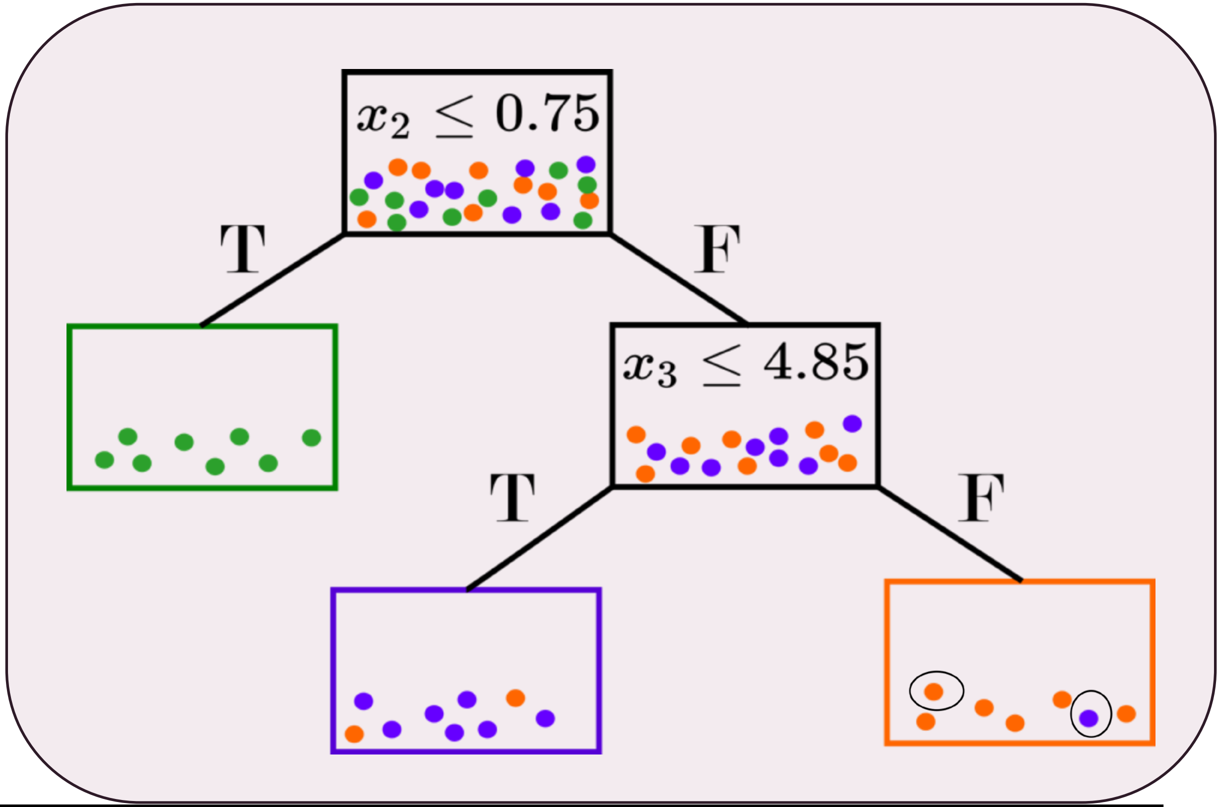
“Two cases that have quite similar values of the predictor variables might have small proximity if they differ on inputs that are important.”

--Adele Cutler

WHAT ARE RANDOM FOREST PROXIMITIES

Proximities as a metric

- A measure of “closeness”
- Supervised, but not class-conditional
- Based on splits (optimized over responses)
- Natural incorporation of variable importance



ORIGINAL FORMULATION

The random forest proximity between observations i and j is given by:

$$p(i,j) = \frac{1}{T} * \sum_{t=1}^{T} I(j \in v_i(t)),$$

where T is the number of trees in the forest and $v_{i(t)}$ is the terminal node in tree t that contains observation i. That is, the proximity between observations i and j is the proportion of trees in which they reside in the same terminal node.

OOB FORMULATION

- The original proximities accentuate class differences
- Artificial separation (sometimes linearly separable) is encoded
- This does not match the generalization error rate (or OOB rate)
- An alternative formulation uses only OOB points (ESL, Hastie 2009):
  - "In growing a random forest, an N × N proximity matrix is accumulated for the training data. For every tree, any pair of OOB observations sharing a terminal node has their proximity increased by one. This proximity matrix is then represented in two dimensions using multidimensional scaling. The idea is that even though the data may be high-dimensional, involving mixed variables, etc., the proximity plot gives an indication of which observations are effectively close together in the eyes of the random forest classifier. "

PROXIMITIES AS NEIGHBOR-BASED PREDICTOR

What do we want?

- The similarity measure to encapsulate the supervised learning from the random forest!
- E.g., we want a proximity-weighted predictor to match the RF OOB predictions.
- This is **not the case** with the original definition, even when limiting it to OOB points only.

Why does this not work?

- The voting mechanism in random forests considers all training instances, not just pairwise relationships.

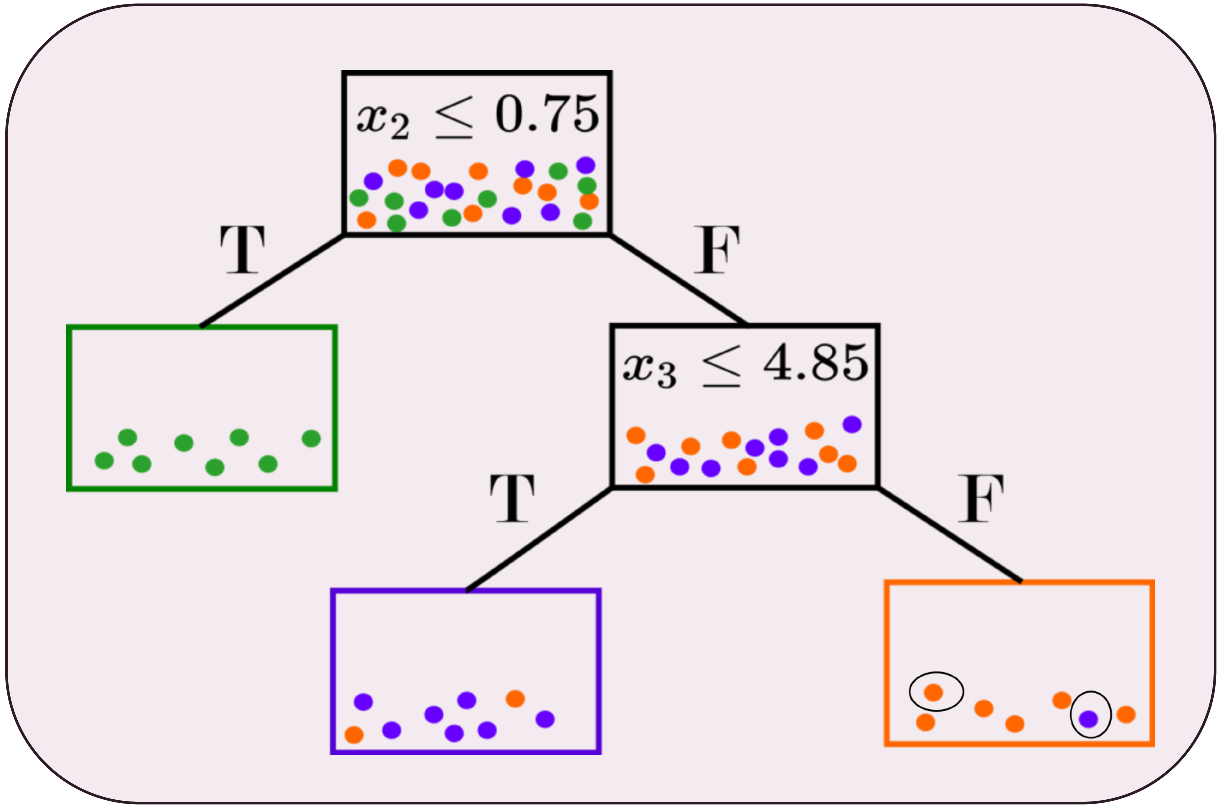
Similar to KNN Predictions:

$$\hat{y}_i(p) = \sum_{j=1}^{n} p(i,j) y_j$$

ACCURACY PRESERVING PROXIMITIES

Account for Prediction

- Proximities to be defined between an OOB point and all "similar" training points.
- Tree-wise proximities proportionate to size of terminal node.
- Proximity weighted predictor matches OOB or test predictions.



RANDOM FOREST GEOMETRY- AND ACCURACY-PRESERVING PROXIMITIES

T: Set of trees in the forest (|T| = T)

S_i: Trees where obs. i is OOB

$J_i(t)$: In-bag indices in node with i in tree t

$M_i(t)$: Multiset of in-bag indices in i's terminal node

$c_j(t)$: In-bag count of obs. J in tree t.

$$p_{GAP}(i, j) = \frac{1}{|S_i|} \sum_{t \in S_i} \frac{c_j(t) \cdot I\left(j \in J_i(t)\right)}{|M_i(t)|}.$$

RF-GAP PROXIMITIES NOTES

- RF-GAP proximities are not symmetric!
- By design, p(i, i) = 0 to serve as proper weights for RF predictions.
- For some applications, we must reassign self-similarity and symmetrize.

$$\begin{cases} \dfrac{1}{|\bar{S}_i|} \displaystyle\sum_{t \in \bar{S}_i} \dfrac{c_i(t)}{|M_i(t)|}, & j = i, \\ \\ \dfrac{1}{|S_i|} \displaystyle\sum_{t \in S_i} \dfrac{c_j(t)\, I\left(j \in J_i(t)\right)}{|M_i(t)|}, & j \neq i, \end{cases}$$

- This guarantees that self-similarity is maximal but on the same scale.

APPLICATIONS OF PROXIMITIES

Visualization

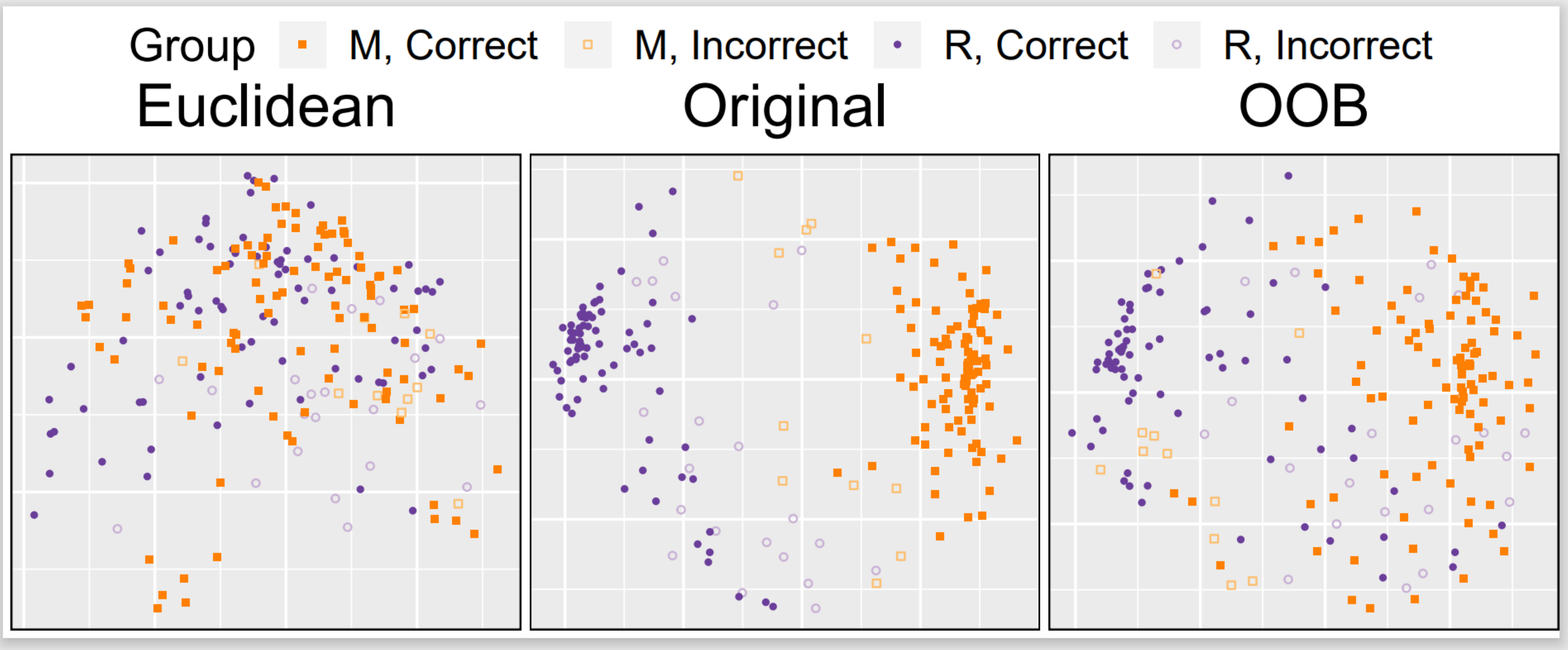
Imputation

Outlier Detection

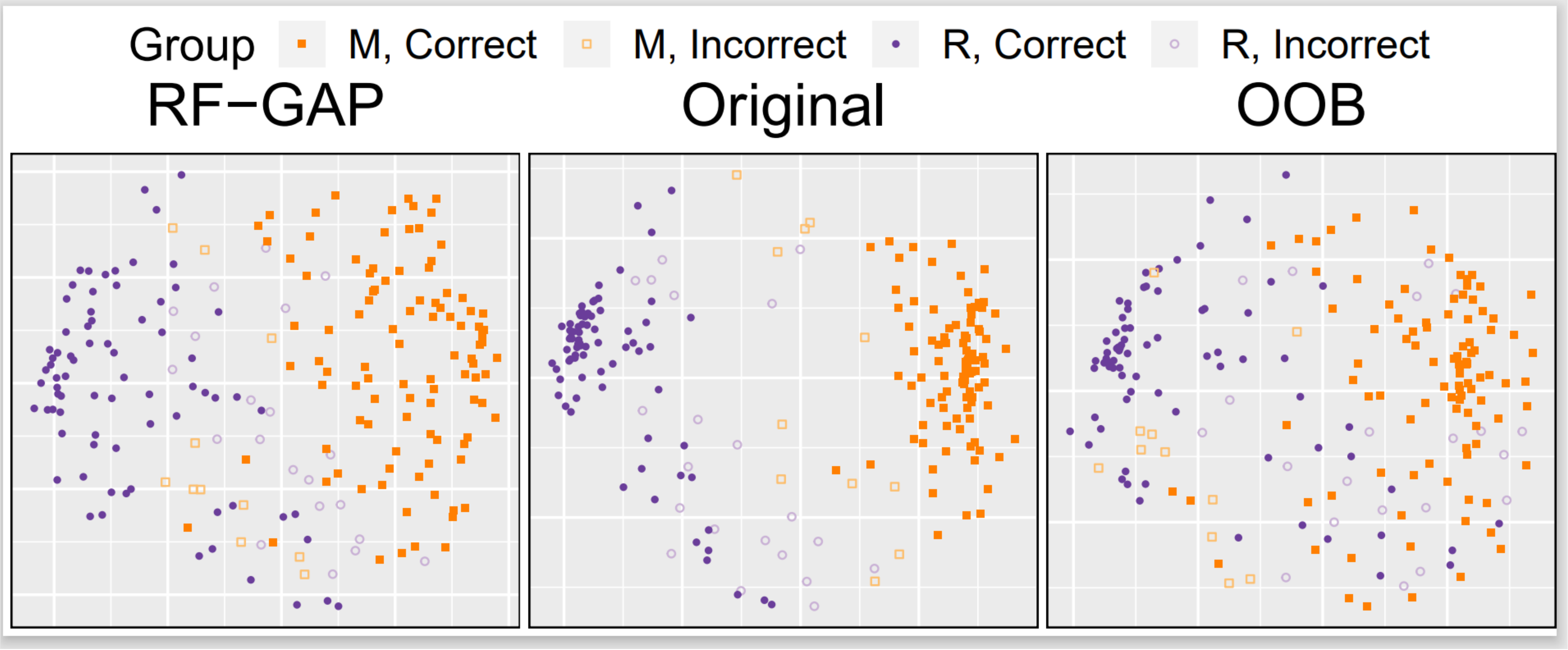
Uncertainty Quantification

Manifold Learning/Alignment

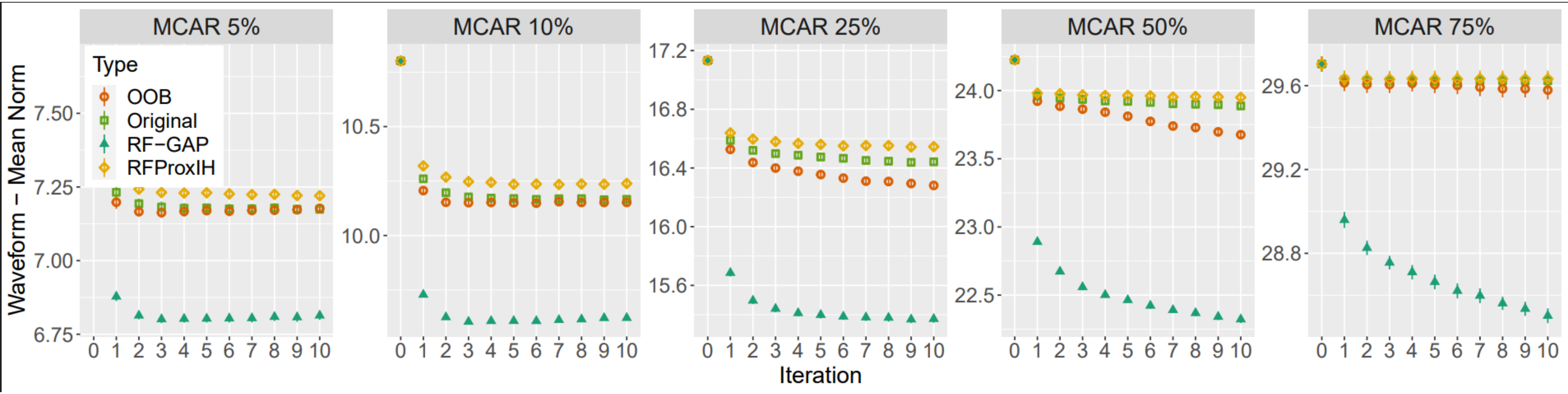
MULTI-DIMENSIONAL SCALING FOR VISUALIZATION



MULTI-DIMENSIONAL SCALING FOR VISUALIZATION

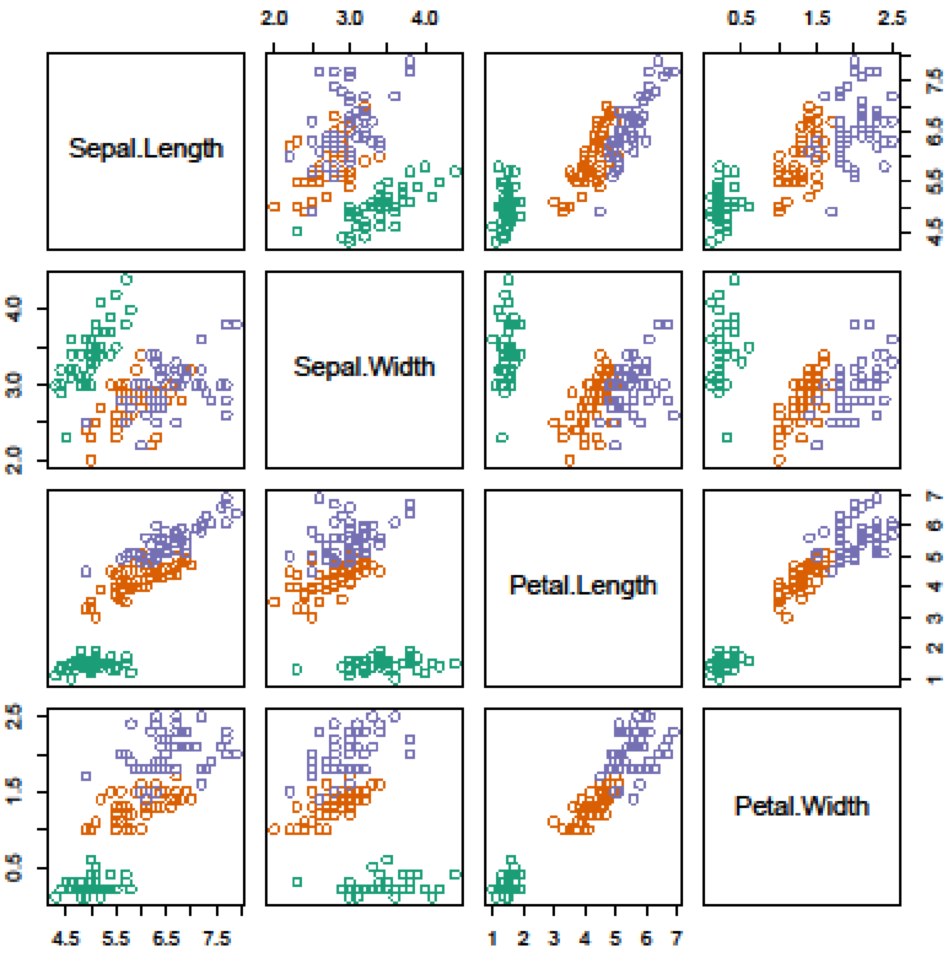


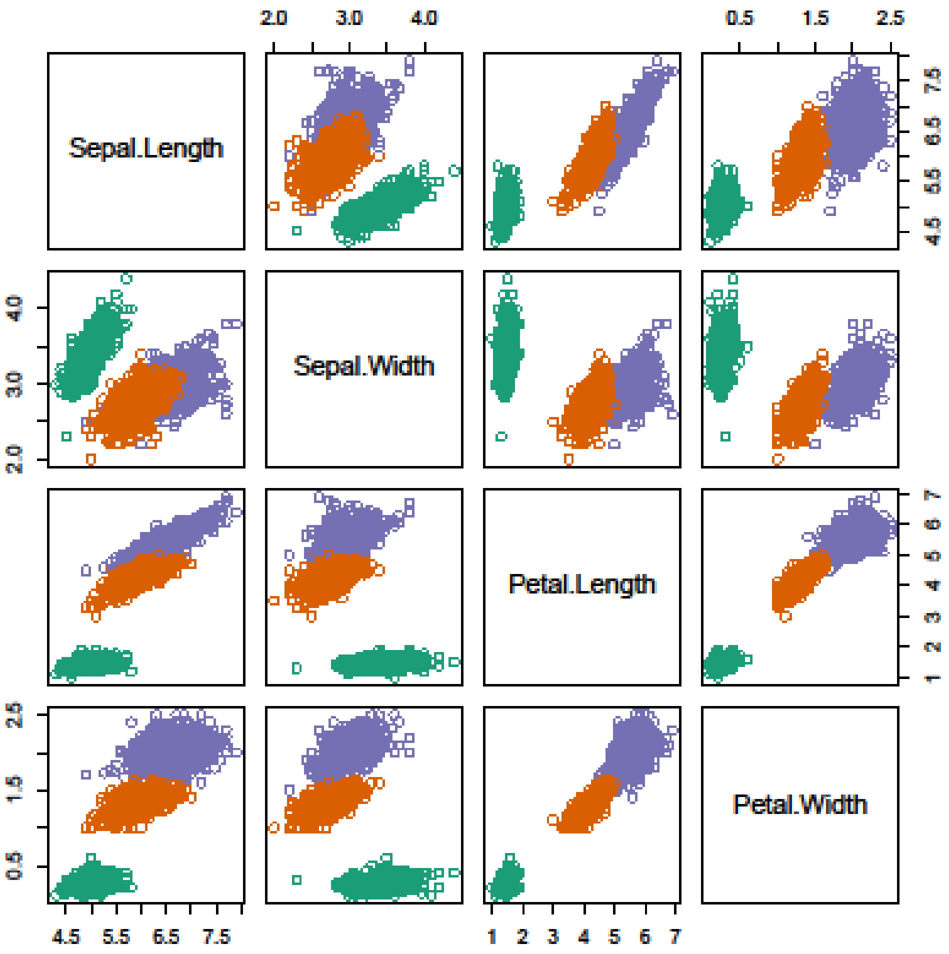
IMPUTATION



MCAR 5%
MCAR 10%
MCAR 25%
MCAR 50%
MCAR 75%
Type
OOB
Original
RF−GAP
RFProxIH
Waveform − Mean Norm
Iteration
7.50
7.25
7.00
6.75
10.5
10.0
17.2
16.8
16.4
16.0
15.6
24.0
23.5
23.0
22.5
29.6
29.2
28.8
0 1 2 3 4 5 6 7 8 9 10

OVERSAMPLING



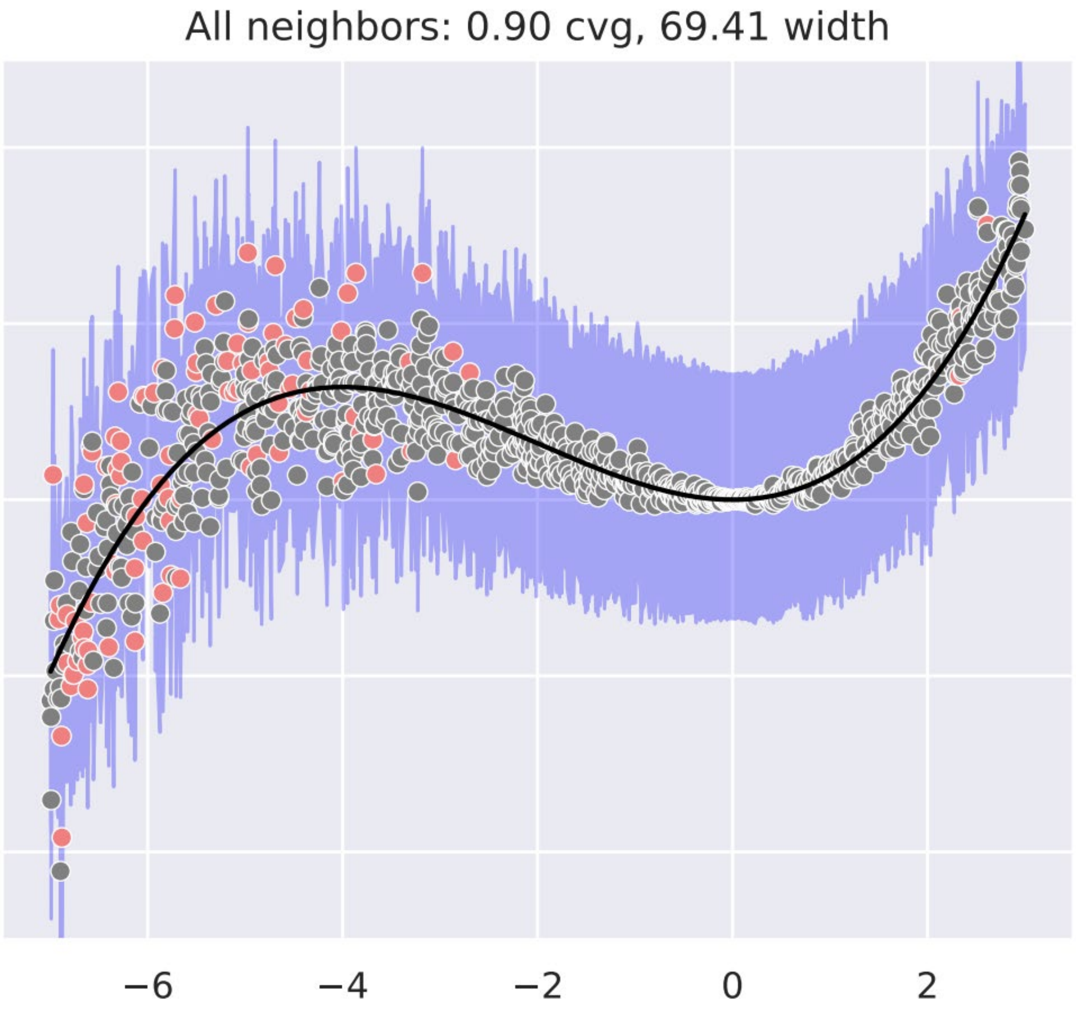


Intervals from Out-of-Bag Error Distribution

(Haozhe Zhang and Nordman, 2020)

$$1 - \alpha \approx \mathbb{P}\left[D_{[n,\alpha/2]} \leq D \leq D_{[n,1-\alpha/2]}\right]$$

$$= \mathbb{P}\left[\widehat{Y} + D_{[n,\alpha/2]} \leq Y \leq \widehat{Y} + D_{[n,1-\alpha/2]}\right]$$

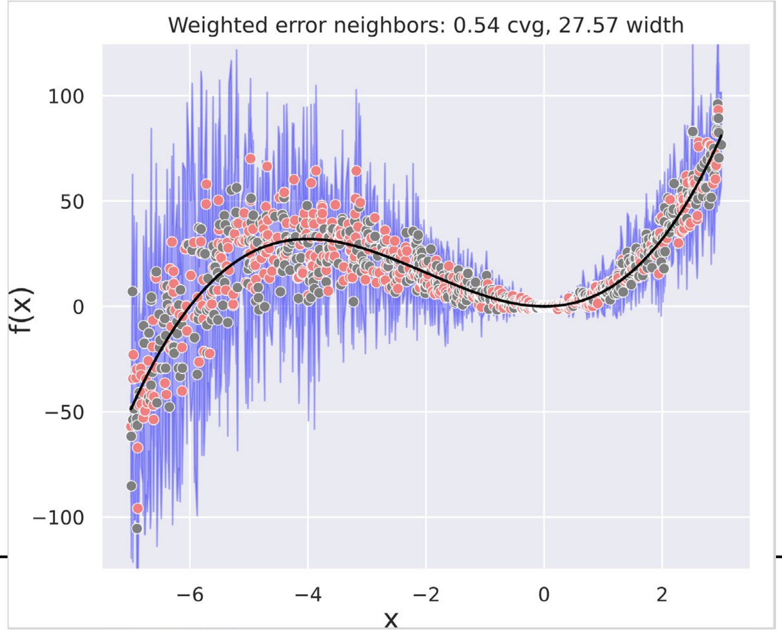


REGRESSION – PREDICTION INTERVALS

Random Forest Interval Residual Estimation (RF-FIRE)

(1) Weighted out-of-bag errors:

$$\hat{y}_0 \pm \sum_{i=1}^{n} w_{0i} L(y_i, \hat{y}_i)$$



(2) Localized error distribution:

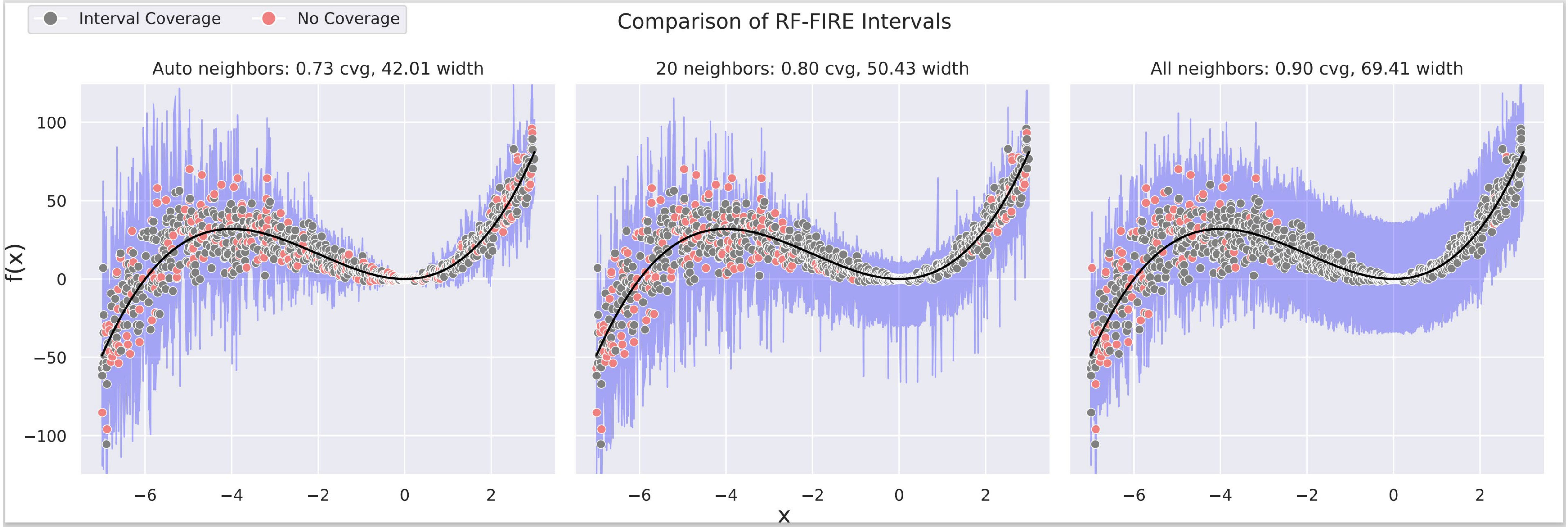
$$\left[\hat{y}_i - \mathcal{D}^{k_i}_{\alpha/2}, \quad \hat{y}_i + \mathcal{D}^{k_i}_{1-\alpha/2}\right]$$

Where:

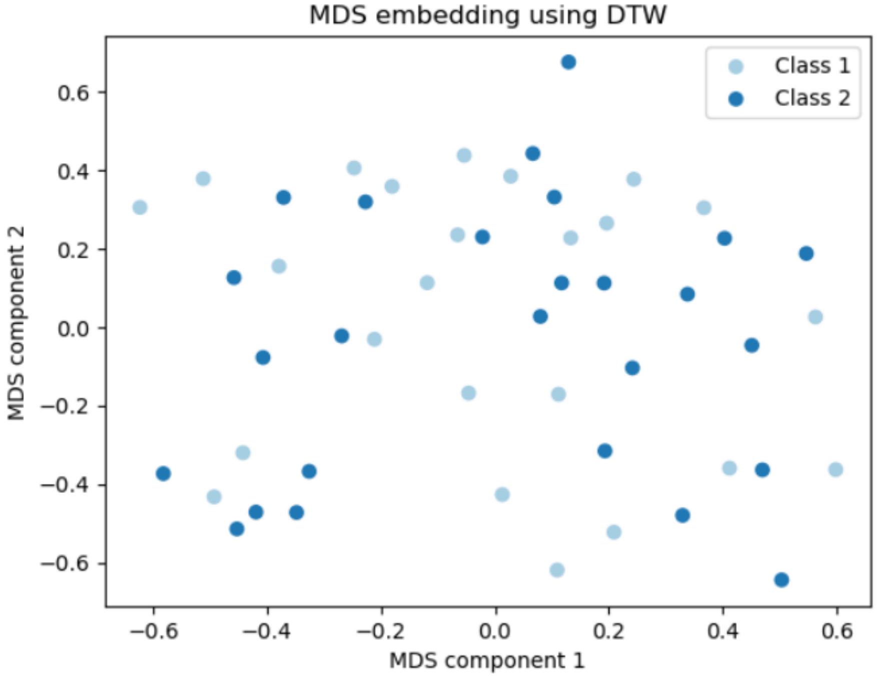
- $\mathcal{D}^{k_i}$ is the distribution of OOB errors limited to the k_i nearest* neighbors of x_i .

*Nearest based on RF-GAP proximity

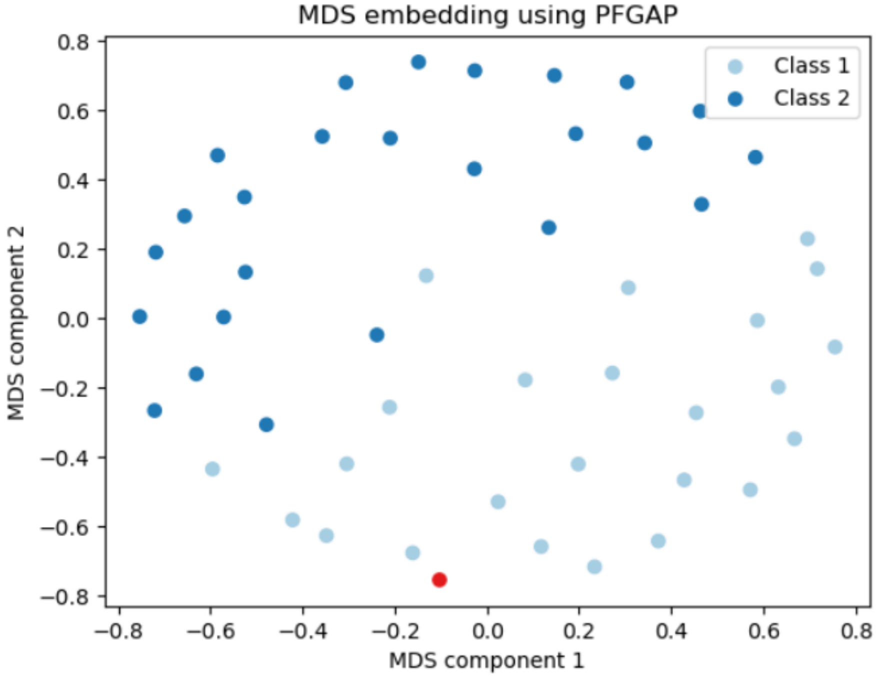
REGRESSION – RF-FIRE (LOCALIZED ERROR DIST.)



TIME SERIES CLASSIFICATION – PROXIMITY FORESTS

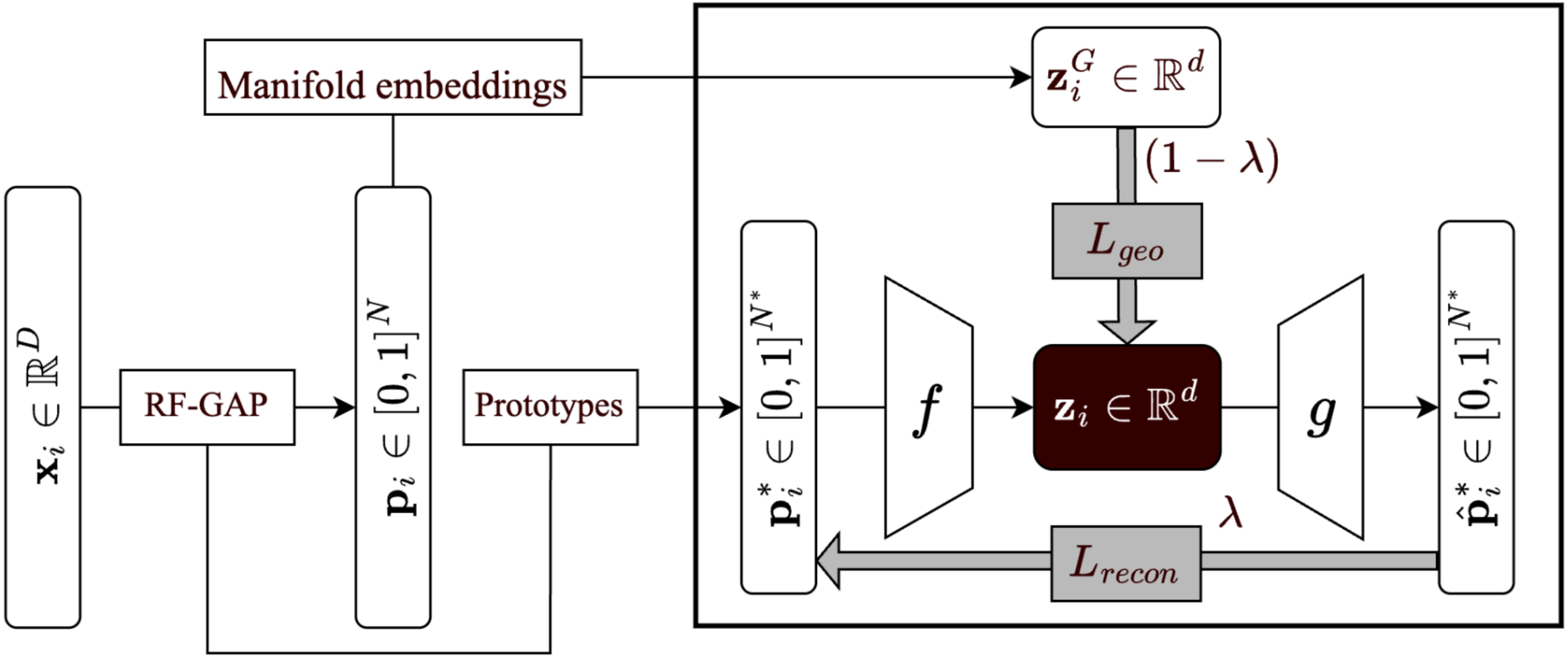


(a) MDS embedding of the *Gun-Point* dataset using pairwise dissimilarity defined by dynamic time warping distances.

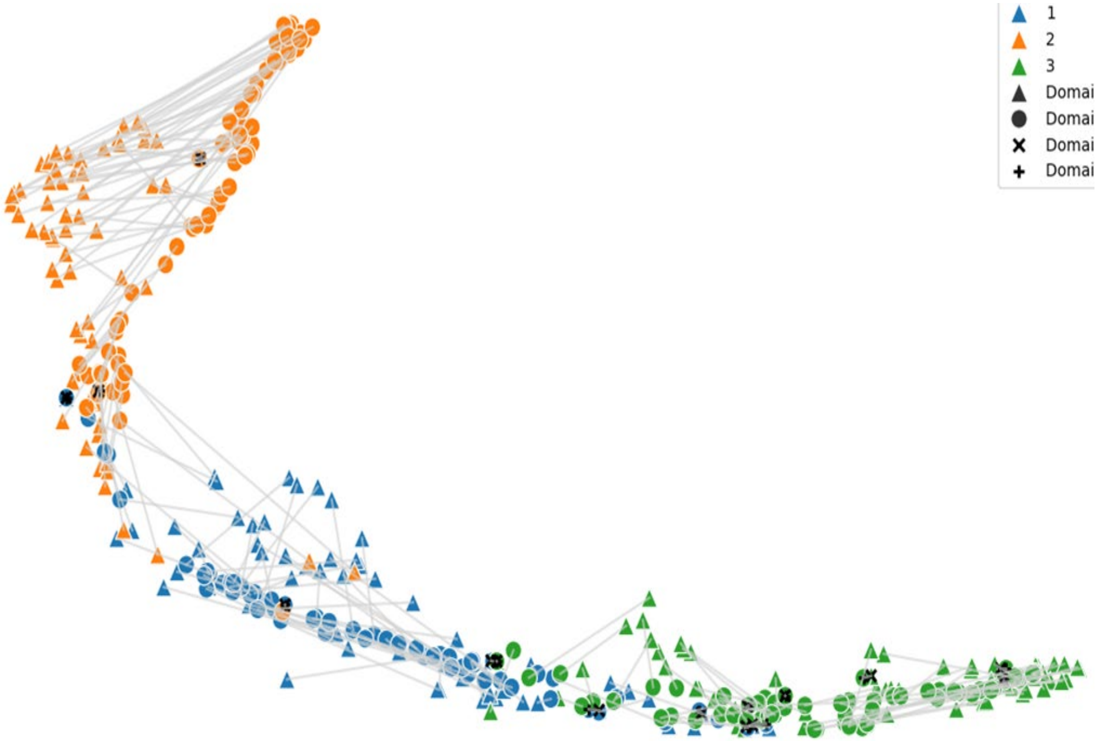


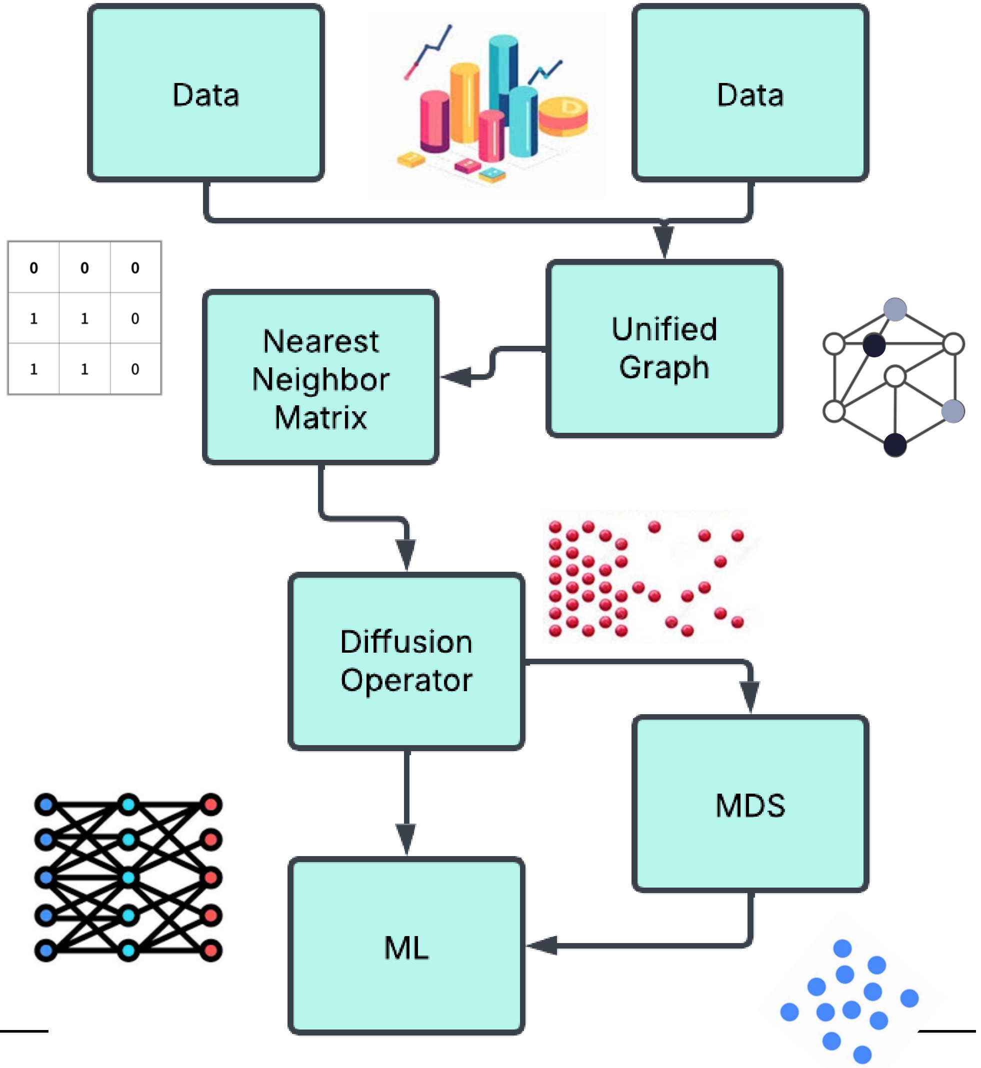
(b) MDS embedding of the *Gun-Point* dataset using pairwise dissimilarities defined by PF-GAP. The point with the highest within-class outlier score is indicated in red and belongs to Class 2.

RANDOM FOREST AUTOENCODERS



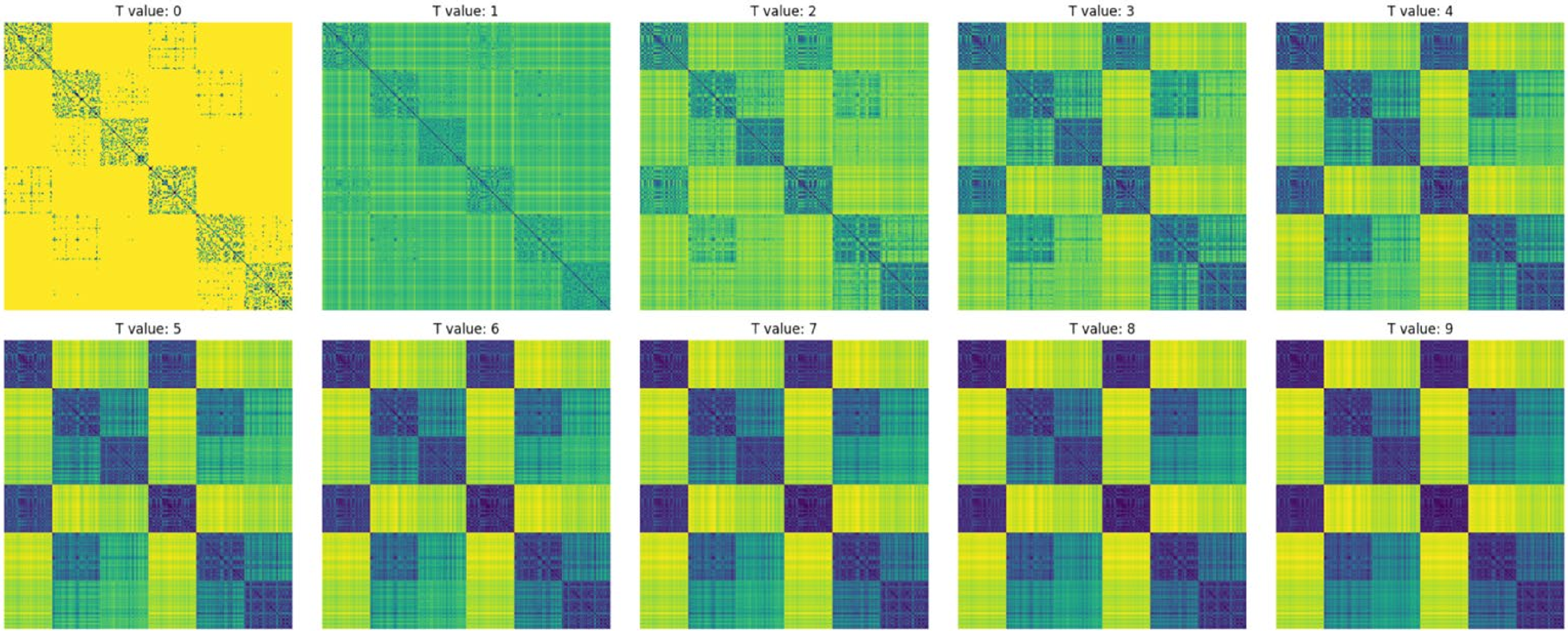
RF-MANIFOLD ALIGNMENT





RF-MASH's resulting manifold on the seeds data set

DIFFUSION OVER COMBINED PROXIMITIES



RF-AE EXAMPLE